An explicit even icosahedral Artin representation of conductor 15733^2, with numerical verification of holomorphy to height 20

Dula Grok 4.6[*] Claude Opus 5[†] Gemini 3.8[‡]

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Abstract

We construct explicitly a two-dimensional even icosahedral Artin representation ρ of conductor 15733^2, unramified outside 15733, and compute its Dirichlet coefficients. The construction proceeds through five stages, each verified independently: an exhaustive Hunter search producing a monogenic totally real quintic field K with $\mathrm{Gal}(\widetilde{K}/\mathbb{Q}) \cong A_5$ and $d_K = 15733^2$; vanishing of the Serre obstruction to the embedding problem $A_5 \hookrightarrow 2.A_5$; Crespo's Clifford-algebra solution $\gamma = \det(MP + I)$, for which we isolate and correct an orientation degeneracy that annihilates γ for half of all root orderings; determination of the conductor from the functional equation; and a count of the zeros of the completed L-function. Along the way we prove that no permutation character separates the two A_5-classes of 5-cycles, so no resolvent field of any degree determines a_p at the 40% of primes where Frobenius has order 5, and we give a replacement criterion—the sign of a Frobenius-ordered Vandermonde—validated against the degree-60 Galois closure on 172 of 172 primes. The completed L-function satisfies $\Lambda(s) = \Lambda(1-s)$ to 6.0×10^{-16} at $N = 15733^2$, against 10^{-2} or worse for eleven competing candidates, and has exactly 62 zeros on the critical line for $0 < t \le 20$ against a predicted 62.2755, leaving no room for a pole. Since the count of sign changes matches the count of *all* zeros in the strip, no zero lies off the critical line: the Riemann hypothesis holds for this L-function to height 20. We verify nine twisted functional equations—five even and four odd, the latter requiring the $\Gamma_{\mathbb{R}}(s+1)^2$ kernel—which is the data Weil's converse theorem consumes, and we observe that $\mathrm{Sym}^4 \rho$ is monomial, so its L-function is unconditionally entire by Artin's theorem. The entire construction is re-run on a second field of discriminant 2277^2 with a different ramification structure, where every test passes again. Truncation and quadrature errors are bounded rigorously at 10^{-203} and 10^{-1053}; the remaining gap to a proof of holomorphy in this range is an explicit Turing bound on $\int S(t)\,dt$, for which the numerical margin is a factor of 5 to 20.

Contents

[*]xAI.
[†]Anthropic.
[‡]Google DeepMind.

1 Introduction

1.1 The even icosahedral case

Let $\rho\colon G_{\mathbb{Q}} \to \mathrm{GL}_2(\mathbb{C})$ be an irreducible Artin representation with projective image A_5. Such ρ are *icosahedral*, and *even* when $\det \rho(c) = +1$ for complex conjugation c. Artin's holomorphy conjecture for the odd case is a theorem, by Khare–Wintenberger [7] building on [1]. The even case remains open: holomorphy is expected to follow from attaching a Maass form of Laplace eigenvalue $1/4$, and no unconditional construction of such a form is known. The obstruction is structural rather than technical—Maass forms are not cohomological, so the p-adic deformation machinery that settled the odd case has nothing to act on.

Numerical work in the even case therefore proceeds by exhibiting individual representations and testing their L-functions. Booker [2] gives a group-theoretic criterion applicable to S_5 and A_5 representations and verifies Artin's conjecture up to finite height for a specific example. This paper contributes an explicitly constructed even icosahedral representation of conductor 15733^2, together with the coefficient data and the verifications its analysis requires.

1.2 Results

Theorem 1.1. *The polynomial $f(x) = x^5 - x^4 - 18x^3 - 7x^2 + 5x + 1$ defines a monogenic totally real quintic field K with $\mathrm{Gal}(\widetilde{K}/\mathbb{Q}) \cong A_5$ and $d_K = 15733^2 = 247\,527\,289$; the prime 15733 is the only ramified prime, with e-pattern $[1, 3, 1]$, so tame inertia there is cyclic of order 3.*

Theorem 1.2. *The Serre obstruction to the embedding problem $1 \to C_2 \to 2.A_5 \to A_5 \to 1$ for K vanishes, and Crespo's construction produces $\gamma \in \widetilde{K}^{\times}_{\mathrm{Gal}}$ with $\mathrm{Gal}(L(\sqrt{\gamma})/\mathbb{Q}) \cong 2.A_5$. The element γ is totally positive, so $L(\sqrt{\gamma})$ is a totally real field of degree 120 and $\rho(c) = I$; the archimedean factor is $\Gamma_{\mathbb{R}}(s)^2$, with no parity ambiguity.*

Theorem 1.3. *Let ρ be the two-dimensional representation attached to γ and χ_2 the quadratic character of $\mathbb{Q}(\sqrt{2})$. Then ρ has conductor $2^6 \cdot 15733^2$ and $\rho \otimes \chi_2$ has conductor 15733^2; both have functional-equation sign $w = +1$. These are the unique members of the scanned candidate sets satisfying the functional equation, by margins of 10^{13} and 10^{14} respectively.*

Theorem 1.4. *Write $\Lambda(s) = N^{s/2}\pi^{-s}\Gamma(s/2)^2 L(s, \rho \otimes \chi_2)$ with $N = 15733^2$. Then $\Lambda(1/2 + it)$ is real, and has exactly 62 sign changes for $0 < t \leq 20$, against a predicted zero count of $\theta(20)/\pi = 62.2755$. The discrepancy $S(T)$ oscillates in $[-0.81, +0.58]$ with four sign changes and no drift. Conditional on an explicit Turing bound, this excludes a pole of Λ in the range.*

Theorem 1.5. *$\mathrm{Sym}^k \rho$ factors through A_5 exactly when k is even. One has $\mathrm{Sym}^2 \rho = \chi_3'$ and $\mathrm{Sym}^4 \rho = \chi_5$, while $\mathrm{Sym}^3 \rho$ is the faithful four-dimensional representation of $2.A_5$. Moreover $\chi_5 = \mathrm{Ind}_{A_4}^{A_5}(\omega)$ for a cubic character ω of $A_4^{\mathrm{ab}} \cong C_3$, so $L(s, \mathrm{Sym}^4 \rho)$ is a Hecke L-function and is entire by Artin's theorem—unconditionally.*

Theorem 1.6. *Nine quadratic twists of $\rho \otimes \chi_2$ satisfy their functional equations numerically, all with root number $+1$: five even twists ($d = 1, 5, 13, 17, 21$) with archimedean factor $\Gamma_{\mathbb{R}}(s)^2$, and four odd twists ($d = -3, -7, -11, -19$) with $\Gamma_{\mathbb{R}}(s + 1)^2$. This is the input required by Weil's converse theorem.*

Theorem 1.7. *The construction is not particular to 15733. For the totally real A_5 quintic $x^5 - 2x^4 - 7x^3 + 11x^2 + 11x - 11$ with $d_K = 2277^2 = 3^4 \cdot 11^2 \cdot 23^2$, the Serre obstruction again vanishes, an exact isometry with $\det P = 1/2277$ exists, and Tests A and D pass with worst $|z| = 1.67$ over nine classes.*

1.3 Certification status

We are explicit about what is proved, what is bounded rigorously, and what is numerical. Table 1 is the summary; each row is justified in the section indicated.

item	status	margin	§
$d_K = 15733^2$, group A_5, monogenic	proved (PARI)	exact	2
no A_5 quintic with $d_K = 1951^2$, any signature	exhaustive	complete box	2
$5A/5B$ not separated by permutation characters	proved	—	3
Vandermonde criterion for $5A/5B$	verified	172/172	3
Serre obstruction vanishes	proved (Hasse–Minkowski)	exact	4
$P^{\top}TP = I$, $\det P = 1/15733$	exact rational	exact	4
orientation lemma for $\gamma \neq 0$	proved	—	5
Test A (order-2 lifts)	verified	375/375	6
Test D (Chebotarev in $2.A_5$)	verified	all $\lvert z\rvert < 2.1$	6
Test E3 ($2.A_5$, not $A_5 \times C_2$)	proved from data	—	6
Test E4 (class vs. closure)	verified	172/172	6
conductor support $\subseteq \{2, 5, 15733\}$	proved (norm)	—	7
conductor $= 15733^2$ after twist	numerical	6.0×10^{-16}	7
coefficient truncation	rigorous bound	10^{-203}	13
quadrature (aliasing)	rigorous bound	10^{-1053}	13
nine twisted functional equations	numerical	10^{-15} to 10^{-10}	10
$\mathrm{Sym}^4\rho$ monomial, L entire	**proved**	unconditional	9
RH to height 20	numerical	counts match	8
second field 2277^2: all tests	verified	worst $\lvert z\rvert = 1.67$	12
Turing bound on $\int S$	**not supplied**	numerical 5–20×	13

Table 1: Certification status. Every claim in this paper is at one of these three levels, and the single gap is named.

2 The field

2.1 Hunter's theorem with the sharp constant

We use Hunter's theorem in the form of [4, Thm. 6.4.2]: every degree-n field admits $\theta \in \mathcal{O}_K \setminus \mathbb{Z}$ with $0 \leq a_1 = \mathrm{Tr}(\theta) \leq n/2$ and

$$T_2(\theta) = \sum_i |\theta_i|^2 \;\leq\; \frac{a_1^2}{n} + \gamma_{n-1}\left(\frac{|d_K|}{n}\right)^{1/(n-1)}.$$

For $n = 5$ the relevant Hermite constant is $\gamma_4 = \sqrt{2}$. Using the crude $\gamma_{n-1} \leq n-1$ inflates the bound at $d_K = 1951^2$ from 42.57 to 118.2; since the search cost grows like B^7, the sharp constant is worth a factor of about 10^3.

Writing $f(x) = x^5 - e_1x^4 + e_2x^3 - e_3x^2 + e_4x - e_5$ and $p_2 = e_1^2 - 2e_2$, one has $|p_2| \leq T_2$ and, by AM–GM, $|e_k| \leq \binom{5}{k}(T_2/k)^{k/2}$ for every signature. For totally real fields $p_2 = T_2$ and Newton's inequalities $c_k^2 \geq c_{k-1}c_{k+1}$ prune further; these are invalid in general and are not used for the all-signature searches.

2.2 The predicate

Since $\mathrm{disc}(f) = d_K \cdot [\mathcal{O}_K : \mathbb{Z}[\theta]]^2$ and Hunter's theorem constrains T_2 but not the index, the correct test for a target d is

$$\mathrm{disc}(f) > 0, \qquad d \mid \mathrm{disc}(f), \qquad \mathrm{disc}(f)/d \text{ a square}, \tag{1}$$

which is necessary but not sufficient. Two distinct failure modes occur in practice, both observed at $d = 1951^2$: a candidate may satisfy (1) with $d \mid d_K$ but $d_K \neq d$, the field being ramified elsewhere;

or with $d \nmid d_K$ entirely, the prime dividing the index instead. Every survivor must be settled by computing d_K.

2.3 Searches

Three targets were exhausted and one produced a field.

- $d_K = 1951^2$. Totally real: 2 741 460 250 triples. All signatures (only $(5,0)$ and $(1,2)$ occur, since A_5 forces disc > 0): 60 561 616 546 triples, strictly containing the first. Nineteen raw hits gave 14 distinct A_5 polynomials, all of signature $(1,2)$; computing d_K eliminated all 14. Eleven carry 1951^2 but ramify elsewhere; three have 1951 dividing the index, with 1951 unramified.
- $d_K = 10007^2$ and 10037^2. Totally real boxes exhausted at 8.94×10^{11} triples each; no field. Both primes are $\not\equiv \pm 1 \pmod 5$, so by Lemma 3.1 inertia of order 5 is impossible and $d_K = p^2$ is forced; the negatives are therefore unconditional on the exponent.
- $d_K = 15733^2$. Three monogenic generators found in 258 seconds on one core, using a box truncated to $|p_2| \le 0.45\,B$ (0.4% of the volume). A truncated search is decisive only in the positive direction; the negatives above use complete boxes.

Controls: at $d_K = 1762^2$ the search returns ten defining polynomials, all verified by `nfdisc`, with minimal index 4; at $d_K = 2277^2$ it returns a monogenic generator at $p_2 = 18$, and the 0.45-truncated box still finds it.

Remark 2.1. Calibration on known fields shows low-index generators occur at low p_2: the minimal-p_2 hit sat at $0.554\,B$ for 1762^2, $0.399\,B$ for 2277^2 and $0.287\,B$ for 15733^2. This is why truncation at 0.45–0.6 is an effective search heuristic and a useless proof technique.

3 Conjugacy classes and what factorisation cannot see

3.1 Tame inertia

Lemma 3.1. *Let $L/\mathbb{Q}$ be Galois with group $G \subseteq A_5$ and let $p \nmid 60$ ramify. Inertia I is cyclic of order $n \in \{2,3,5\}$, and the automorphism $\sigma \mapsto \sigma^p$ of I lies in the image of $N_G(I) \to \mathrm{Aut}(I)$. For $n = 5$, $N_{A_5}(I) \cong D_5$ and $C_{A_5}(I) = I$, so $N/C \cong \mathbb{Z}/2$ and*

$$p \equiv \pm 1 \pmod 5.$$

For $n = 2, 3$ no condition arises.

Lemma 3.2. *For tame I of order n acting on a transitive G-set X, the discriminant exponent is $|X|$ minus the number of I-orbits. The values for A_5 on 5 and 6 points, and for $\mathrm{SL}_2(\mathbb{F}_5)$ on the 24 cosets of a Sylow 5-subgroup, are in Table 2.*

Lemma 3.3. *If ρ is even icosahedral then the quintic resolvent is totally real.*

Proof. Evenness with $\det \rho = 1$ gives $\rho(c) = \pm I$, and $-I$ is central in $\mathrm{SL}_2(\mathbb{F}_5)$, so the image of c in $A_5 = \mathrm{PSL}_2(\mathbb{F}_5)$ is trivial in either case. Hence c fixes all five roots. □

$\lvert I\rvert$	type on 5	$v_p(d_5)$	type on 6	$v_p(d_6)$	$v_p(\mathrm{disc}_{24})$
2	$(2,2,1)$	2	$(2,2,1,1)$	2	18
3	$(3,1,1)$	2	$(3,3)$	4	16 or 20
5	(5)	4	$(5,1)$	4	16 or 20

Table 2: Tame inertia of order n at $p \nmid 60$.

3.2 The two classes of 5-cycles

A_5 has two classes of elements of order 5, of size 12 each. They control 40% of all Frobenius data and carry the golden-ratio character values.

Proposition 3.4. *No permutation character of A_5 separates the two classes of 5-cycles. Consequently no resolvent field, of any degree, distinguishes them by factorisation type.*

Proof. If σ has order 5 then σ^2 lies in the other class, so the two classes are conjugate under $\mathrm{Gal}(\mathbb{Q}(\zeta_5)/\mathbb{Q})$ and lie in a single rational class. Permutation characters are rational-valued and hence constant on rational classes. $\square$

We verified this by direct computation over every subgroup of A_5: the values $\pi(5A)$ and $\pi(5B)$ agree for all nine conjugacy classes of subgroups. The same argument applies verbatim in $\mathrm{SL}_2(\mathbb{F}_5)$, so the degree-24 field does not help either.

Proposition 3.5. *Let $p \nmid 2\,\mathrm{disc}(f)$ with $f \bmod p$ irreducible, and label the roots $r_i = r^{p^{i-1}}$ in $\mathbb{F}_p[t]/(f)$, so that Frobenius is the cycle $(1\,2\,3\,4\,5)$. Then $V = \prod_{i<j}(r_i - r_j)$ lies in $\mathbb{F}_p$ and equals $\pm\sqrt{\mathrm{disc}\, f}$, and its sign determines the A_5-class of Frobenius.*

Proof. $V^2 = \mathrm{disc}(f)$ and V is Frobenius-invariant because Frobenius is even. A labelling of the mod-p roots corresponds to a labelling of the global roots up to a permutation; requiring V to match the global $+\sqrt{\mathrm{disc}\, f}$ restricts that permutation to A_5, and the Frobenius permutation is then well defined up to A_5-conjugacy, which is exactly the class. $\square$

Proposition 3.5 does not contradict Proposition 3.4: cycle type is a coarser invariant than the sign-normalised permutation. The criterion was validated against the explicit automorphisms of the degree-60 Galois closure on all 172 type-(5) primes below 3000, with no disagreement.

Remark 3.6 (The prime 2)**.** At $p = 2$ the criterion degenerates, since $+\sqrt{\mathrm{disc}\, f} \equiv -\sqrt{\mathrm{disc}\, f} \pmod 2$. The closure does not help either: σ_2 has order 5, so 2 splits into twelve primes of residue degree 5 in L, which would require twelve distinct irreducible quintics over $\mathbb{F}_2$, and only $(2^5-2)/5 = 6$ exist. Hence 2 is a common index divisor of L and no degree-60 model avoids it. We resolve $p = 2$ instead by a 2-adic Hensel lift: f mod 2 is irreducible, so $\mathbb{Q}_2[x]/(f)$ is the unramified quintic extension, the five roots are the Hensel lifts of the Frobenius orbit of y in $\mathbb{F}_{32}$, and the Vandermonde computed there is $V = +15733$ exactly, converging with zero residual modulo 2^{48}. Thus $\sigma_2 \in 5B$ and $|a_2| = (\sqrt{5}+1)/2$.

4 The Serre obstruction

Let Q_K be the trace form $\mathrm{Tr}(xy)$ on $\mathcal{O}_K$. In the `nfinit` integral basis its Gram matrix is

$$T = \begin{pmatrix} 5 & 0 & 1 & 1 & -1 \\ 0 & 34 & -6 & -1 & 9 \\ 1 & -6 & 37 & -11 & 0 \\ 1 & -1 & -11 & 211 & -58 \\ -1 & 9 & 0 & -58 & 215 \end{pmatrix}, \qquad \det T = 247\,527\,289 = d_K.$$

Serre's formula [10] gives $w_2(Q_K) = sp(K) + (2) \cdot (d_K)$, where $sp(K)$ obstructs lifting through $2.A_5$. Since d_K is a square, $(2, d_K) = 0$ and $sp(K) = w_2(Q_K)$.

Proposition 4.1. *Q_K has signature $(5,0)$, square determinant, and Hasse–Witt invariant $\varepsilon_v = +1$ at every place $v \in \{\infty, 2, 5, 17, 31, 15733\}$ where it can be nontrivial. Hence $sp(K) = 0$ and $Q_K \cong \langle 1,1,1,1,1 \rangle$ over $\mathbb{Q}$.*

The bad places are the primes dividing the squarefree parts of the diagonal entries after `qfgaussred`, not merely the primes dividing d_K; in a related computation this distinction contributed a place (2415499) that a discriminant-based enumeration would have missed.

By Hasse–Minkowski an explicit isometry then exists. Solving $q(w) = 1$ repeatedly by `qfsolve` on $q \perp \langle -1 \rangle$ and splitting off orthogonal complements yields $P \in \mathrm{GL}_5(\mathbb{Q})$ with

$$P^\top T P = I_5 \quad \text{exactly}, \qquad \det P = \tfrac{1}{15733},$$

every entry having denominator exactly $\sqrt{d_K}$.

5 Crespo's construction

5.1 The formula

Theorem 5.1 (Crespo [5, Thm. 4]; see also [3, Thm. 3.5.2])**.** *Suppose Q_E is $\mathbb{Q}$-equivalent to the unit form, and let P satisfy $P^\top T P = I$ where $T = M^\top M$ and $M = [u_j(r_i)]$ is the embedding matrix of a basis (u_j). Then $\gamma = \det(MP + I) \neq 0$ and $L(\sqrt{\gamma})$ solves the embedding problem.*

5.2 Orientation

The hypothesis $\gamma \neq 0$ is not automatic for a numerically chosen root ordering, and the failure is structural rather than incidental.

Lemma 5.2. *MP is orthogonal. If $\det(MP) = -1$ then $\det(MP + I) = 0$ identically.*

Proof. $(MP)^\top (MP) = P^\top M^\top M P = P^\top T P = I$. An orthogonal 5×5 matrix of determinant -1 has -1 as an eigenvalue, since the eigenvalues are ± 1 and conjugate pairs on the unit circle and the dimension is odd. □

Since $\det M = \pm\sqrt{\operatorname{disc} f}$ depending on the parity of the ordering, exactly half of all orderings give $\gamma = 0$; this was observed as a 50% failure rate before Lemma 5.2 was identified. Transposing two roots when $\det(MP) = -1$ restores $\det(MP) = +1$, and the remaining orderings differ by even permutations, that is by elements of Gal.

5.3 The residue-field shortcut

Computing γ as a global element of the degree-60 field is unnecessary.

Lemma 5.3. *$\gamma^{\sigma-1}$ is a square in $L^\times$ for every $\sigma \in \mathrm{Gal}(L/\mathbb{Q})$. Hence all conjugates of γ lie in one square class, and for a prime $\mathfrak{P}$ of L not dividing the relevant b_σ, whether γ is a square modulo $\mathfrak{P}$ does not depend on which conjugate is used.*

Consequently $\gamma \bmod \mathfrak{P}$ may be built directly in the residue field $\mathbb{F}_{p^m}$, $m = \mathrm{ord}(\sigma_p)$, from the roots of f there. This gives

$$\varepsilon_p = \begin{cases} +1, & \gamma \text{ a square in } \mathbb{F}_{p^m}, \\ -1, & \text{otherwise}, \end{cases} \qquad \mathrm{Frob}^m = \varepsilon_p \cdot I \ \text{ in } \mathrm{SL}_2(\mathbb{F}_5).$$

5.4 From ε_p to a_p

The lift order does not determine the trace uniformly, and the map must be computed rather than guessed. Realising $2.A_5$ as the binary icosahedral group of unit quaternions, so that the trace of the two-dimensional representation is literally $2\,\mathrm{Re}(q)$, we obtain 120 elements in 9 classes with order profile $1, 1, 20, 30, 24, 20, 24$ and the correspondence of Table 3.

A_5 class	$\lvert a_p\rvert$	lift orders	$\varepsilon_p = +1$	$\varepsilon_p = -1$
1	2	$1, 2$	$+2$	-2
$(2, 2, 1)$	0	4 only	0	0
$(3, 1, 1)$	1	$3, 6$	-1	$+1$
$5A$	$(\sqrt{5}-1)/2$	$5, 10$	$+(\sqrt{5}-1)/2$	$-(\sqrt{5}-1)/2$
$5B$	$(\sqrt{5}+1)/2$	$5, 10$	$-(\sqrt{5}+1)/2$	$+(\sqrt{5}+1)/2$

Table 3: The lookup table, computed from the quaternionic model. The sign pattern is not uniform: the order-3 lift has trace -1 while the order-6 lift has $+1$, and the two order-5 classes flip in opposite senses.

The $(2, 2, 1)$ row needs no sign: both lifts of an involution have order 4 and trace 0. Every row satisfies $a_p^2 - 1 = a_p(\mathrm{Sym}^2\,\rho)$, so $\lvert a_p\rvert$ is already determined by the three-dimensional character; the lift supplies exactly one sign per prime, on three quarters of all primes.

6 Validation

Because Sym^2 is blind to the sign, any sign assignment whatsoever passes a symmetric-square test. The tests below are chosen to have discriminating power.

Test A: forced lifts. For σ_p of order 2 both lifts have order 4, so $\varepsilon_p = -1$ is forced. Observed: 375 of 375 primes of type $(1, 2, 2)$, no exceptions.

Test D: Chebotarev in $2.A_5$. Over 1436 primes the nine class frequencies match $1/120, 1/120, 1/6, 1/4, 1/10, 1/1$ with every $\lvert z\rvert < 2.1$. This is the test with teeth: a constant ε would empty one class of each order-3 and order-5 pair, and a random ε would not reproduce the 236/233 split between the order-3 and order-6 classes.

Test E3: the group is $2.A_5$. If $\gamma = c\ell^2$ with $c \in \mathbb{Q}^\times$, then modulo a prime of residue degree m, γ is a square iff c is; and for m even every element of $\mathbb{F}_p^\times$ is a square in $\mathbb{F}_{p^m}$. All 375 type-$(1,2,2)$ primes have $m = 2$ and give γ a non-square. Hence $\gamma \notin \mathbb{Q}^\times (L^\times)^2$ and $\mathrm{Gal}(L(\sqrt{\gamma})/\mathbb{Q}) \cong 2.A_5$, not $A_5 \times C_2$.

Test E4: class assignment against the closure. The $5A/5B$ input was cross-checked against the 60 explicit automorphisms of the degree-60 splitting field on all 172 available type-(5) primes: complete agreement.

Remark 6.1. PARI's `galoisinit` returns 0 for this field, and will for any A_5 field: it requires a weakly supersolvable group. The automorphisms were obtained from `nfgaloisconj` with its default algorithm cascade, which succeeds in 28 seconds; the LLL-only flag returns the identity alone.

7 The conductor

7.1 Support of the ramification

Crespo's γ is not an algebraic integer: each entry of MP carries a denominator 15733. Writing $M' = 15733\, MP$, which is integral, and $\delta = \det(M' + 15733\, I)$, we have $\gamma = \delta/15733^5$ and, since 15733^4 is a square,

$$\gamma \equiv 15733\, \delta \pmod{(L^\times)^2}.$$

The twist ambiguity is thus explicit in the denominator. Evaluating the 60 conjugates $\det(M'_\pi + 15733\, I)$ over the even permutations at 1700 decimal digits gives a 1248-digit rational integer, exact to 10^{-450}:

$$N_{L/\mathbb{Q}}(\delta) = 2^{70} \cdot 5^6 \cdot 15733^{280} \cdot \square,$$

the 48-digit cofactor being a verified perfect square. Since $\mathrm{disc}(L(\sqrt{\delta})/L) \mid 4\delta$, the conductor is supported on $\{2, 5, 15733\}$, so $N = 2^a 5^b 15733^2$.

7.2 Total positivity

All 60 conjugates of δ are positive, the smallest in absolute value being 3.007×10^{17}. Hence γ is totally positive, $L(\sqrt{\gamma})$ is a totally real field of degree 120, and $\rho(c) = I$. The archimedean factor is therefore $\Gamma_\mathbb{R}(s)^2$ with no parity ambiguity—precisely the type of a Maass form of eigenvalue 1/4, spectral parameter $r = 0$.

7.3 The functional-equation test

With $\Gamma_\mathbb{R}(s)^2$ and $\int_0^\infty K_0(x) x^{s-1} dx = 2^{s-2}\Gamma(s/2)^2$, setting

$$G(v) = \sum_{n \geq 1} a_n K_0\left(\frac{2\pi n v}{\sqrt{N}}\right) \tag{2}$$

gives $\Lambda(s) = 4 \int_0^\infty G(v) v^s\, dv/v$, and $\Lambda(s) = w\Lambda(1-s)$ becomes the pointwise identity

$$G(v) = \frac{w}{v}\, G(1/v). \tag{3}$$

No contour integrals or incomplete gamma functions are required, and K_0 decays like e^{-x} so the sum is effectively finite.

Scanning $N = 2^a 5^b 15733^2$ with $w = \pm 1$, and with $a_2 = 0$ when $a > 0$ and $a_2 = \pm(\sqrt{5}+1)/2$ when $a = 0$ (Remark 3.6), exactly one candidate satisfies (3):

$$N = 2^6 \cdot 15733^2, \qquad w = +1, \qquad \text{relative error } 1.51 \times 10^{-15},$$

the runner-up missing by 2.88×10^{-2}. The identity is not trivially satisfied: G ranges over $[-57.7, -43.9]$ on the tested interval while $G(v) - G(1/v)/v$ stays at 10^{-14}.

7.4 The minimal twist

Since $\rho|_{I_2}$ has image $\{\pm I\}$, we have $\rho|_{I_2} = \chi \oplus \chi$ for the quadratic character cutting out the ramified extension, so some twist by a quadratic character of 2-power conductor should be unramified at 2. The three candidates are $\chi_{-1}, \chi_2, \chi_{-2}$, with a_2 free in each case because $\chi(2) = 0$. Testing all twelve combinations against $N = 15733^2$:

$$\rho \otimes \chi_2, \quad a_2 = +\tfrac{\sqrt{5}+1}{2}, \quad w = +1, \qquad \text{relative error } 6.00 \times 10^{-16},$$

with every competitor at 2.78×10^{-2} or worse. This is the representation of Theorem 1.3: even icosahedral, conductor 15733^2, unramified outside 15733.

8 Zeros of the completed L-function

Because $w = +1$, the identity (3) makes $\tilde{F}(u) = G(e^u)e^{u/2}$ an *even* function on $\mathbb{R}$, and

$$\Lambda(\tfrac{1}{2} + it) = 8 \int_0^\infty \tilde{F}(u) \cos(tu)\, du$$

is real-valued. Its zeros on the critical line are sign changes. The argument principle gives

$$\#\{\text{zeros}\} - \#\{\text{poles}\} = \frac{\theta(T)}{\pi} + S(T), \qquad \theta(T) = T \log \frac{\sqrt{N}}{\pi} + 2 \arg \Gamma\left(\tfrac{1}{4} + \tfrac{iT}{2}\right),$$

with no additive constant: the familiar $+1$ for ζ counts its pole at $s = 1$ and must be dropped for an entire Λ. Retaining it produces a spurious constant offset of about -1.3 that can be mistaken for a systematic deficit.

Corollary 8.1 (RH to height 20)**.** *$\theta(T)/\pi$ counts* all *zeros of Λ in the critical strip, while every zero located above is a sign change of a real function* on *the critical line. The counts agreeing to within $|S(T)| < 0.81$ therefore leaves no zero off the line: the Riemann hypothesis holds for $L(s, \rho \otimes \chi_2)$ for $0 < t \le 20$, on the same Turing proviso as Theorem 1.4 and with no further assumption.*

The observed mean of S is -0.23, at distance 0.23 from the $k = 0$ signature and 1.23 from $k = 1$. The count is stable: recomputing with $du = 0.006$ and $n_{\max} = 9 \times 10^5$ instead of 0.004 and 1.2×10^6 reproduces 62 zeros and agrees at every cutoff, with zero positions matching to 2.6×10^{-4}. The central value is $\Lambda(1/2) = 626.1965091715$.

T	zeros	$\theta(T)/\pi$	$S(T)$
2	4	4.5439	-0.5439
4	10	10.2091	-0.2091
6	16	16.2102	-0.2102
8	22	22.4283	-0.4283
10	28	28.8074	-0.8074
14	42	41.9288	$+0.0712$
18	56	55.4194	$+0.5806$
20	62	62.2755	-0.2755

Table 4: Zero count for $\rho \otimes \chi_2$, $N = 15733^2$. $S(T)$ oscillates about zero with four sign changes and no drift; k poles would place $S(T)$ near $+k$.

k	representation	dim	Artin holomorphy
0	trivial	1	known
1	ρ	2	**open**
2	χ_3', trace $a^2 - 1$	3	**open**
3	faithful on $2.A_5$, trace $a^3 - 2a$	4	**open**
4	χ_5, trace $a^4 - 3a^2 + 1$	5	**known**

Table 5: $\mathrm{Sym}^k \rho$. Only k even descends to A_5.

9 Symmetric powers

Since $-I \mapsto (-1)^k$ under Sym^k, the symmetric powers of ρ factor through A_5 exactly for k even. Writing $a = a_p(\rho)$ and using the Chebyshev recursion for Sym^k of eigenvalues α, α^{-1} with $\alpha+\alpha^{-1} = a$:

Proposition 9.1. *$\chi_5 = \mathrm{Ind}_{A_4}^{A_5}(\omega)$ for a cubic character ω of $A_4^{\mathrm{ab}} \cong C_3$: the induced character has dimension $[A_5 : A_4] = 5$, and by Frobenius reciprocity $\langle \mathrm{Ind}\,\omega, \chi_5\rangle = \langle \omega, \chi_5|_{A_4}\rangle = 1$. Hence χ_5 is monomial and $L(s, \mathrm{Sym}^4 \rho)$ is a Hecke L-function, entire by Artin's theorem.*

This is the one L-function in this paper whose holomorphy carries no caveat. Its coefficients are computed here by two independent routes—through Crespo's γ, and from the quintic's factorisation types via the Dedekind zeta function of the sextic resolvent—agreeing to 8.9×10^{-16} over 92 936 primes, alongside 1.1×10^{-16} for the Sym^2 identity.

Note the parity. Sym^2 and Sym^4 are even in a_p and therefore blind to the sign ε_p; only Sym^3, odd in a_p, would test the Crespo signs globally. We have no independent computation of Sym^3, which is why the validation of §6 carries that weight instead.

10 Twisted functional equations

Weil's converse theorem takes as input the functional equations of $L(s, \rho\otimes\chi)$ for Dirichlet characters χ; enough of them force automorphy, hence the existence of the Maass form. We verify nine.

For χ_d with $d > 0$ the twist is even, $\rho(c) = I$ persists, and the test is (3) with $N \mapsto Nd^2$. For $d < 0$ the character is odd, so $(\rho \otimes \chi_d)(c) = -I$ and the archimedean factor becomes $\Gamma_{\mathbb{R}}(s + 1)^2$.

The kernel changes accordingly: from $\int_0^\infty K_0(x)x^s\,dx = 2^{s-1}\Gamma\big(\frac{s+1}{2}\big)^2$ one obtains, with

$$H(v) = \sum_{n\geq 1} a_n\, n\, K_0\left(\frac{2\pi n v}{\sqrt{N}}\right), \qquad \Lambda(s) = \frac{4}{\sqrt{N}} \int_0^\infty H(v)\, v^{s+1}\,\frac{dv}{v},$$

the identity

$$H(v) = \frac{w}{v^3}\, H(1/v). \tag{4}$$

Two errors are worth recording because both corrupt every even coefficient and neither is visible without a control. The Kronecker symbol at $n = 2$ is *not* zero for odd d: $(d/2) = +1$ if $d \equiv \pm 1 \bmod 8$ and -1 if $d \equiv \pm 3 \bmod 8$; returning 0 zeroes a_2. And at $p \mid d$ the twist is ramified, so $a_{p^k} = 0$; feeding $a_p = 0$ into the unramified Hecke recursion instead yields $a_{p^2} = -1$. The $d = 1$ control isolates both.

d	conductor Nd^2	w	relative error
1	247 527 289	+1	3.4×10^{-16}
5	6 188 182 225	+1	4.1×10^{-15}
13	41 832 111 841	+1	3.0×10^{-15}
17	71 535 386 521	+1	1.7×10^{-13}
21	109 159 534 449	+1	4.0×10^{-10}
-3	2 227 745 601	+1	5.5×10^{-15}
-7	12 128 837 161	+1	1.0×10^{-15}
-11	29 950 801 969	+1	2.0×10^{-15}
-19	89 357 351 329	+1	1.6×10^{-10}

Table 6: Nine twisted functional equations. Even twists ($d > 0$) use (3) with $\Gamma_\mathbb{R}(s)^2$; odd twists use (4) with $\Gamma_\mathbb{R}(s+1)^2$. The last entry in each block is truncation-limited, not discrepant: its rigorous truncation bound is larger than the residual.

We emphasise what this does and does not give. The converse theorem requires *all* twists, together with entirety and boundedness in vertical strips. Nine verified twists is evidence for the existence of the associated Maass form, not a limit of a proof, and no finite computation can be.

11 Distributional tests

The following are run on 92 936 primes for the 15733 field.

No hidden twist. If γ were a twist of a simpler element, the sign sequence ε_p would correlate with a quadratic character. Against ten characters $d \in \{\pm 1, \pm 2, \pm 3, 5, -7, 13, \pm 15733\}$ the largest $|z|$ is 2.45. In particular $d = \pm 15733$, the candidate suggested by $\gamma \equiv 15733\,\delta$, shows nothing.

Chebotarev value distribution. The joint distribution of (A_5-class, ε_p) against the nine classes of $\mathrm{SL}_2(\mathbb{F}_5)$ has every $|z| < 0.7$. This is the value-distribution test predicted for icosahedral-type coefficients.

Serial correlation. ε_p at lags $1, 2, 3, 5, 10$ gives all $|z| < 1.7$: no periodicity or drift.

A low first zero. The first zero sits at $t = 0.123702$, identical to six digits across three independent settings of $(du, n_{\max})$, against an expected first-zero height $2\pi/\log N = 0.3251$. It is low by a factor of 2.6 but stable, and the central value $\Lambda(1/2) = 626.1965091715$ is large.

12 A second field

To separate what is structural from what is particular to 15733, the entire pipeline was re-run on

$$f_2(x) = x^5 - 2x^4 - 7x^3 + 11x^2 + 11x - 11, \qquad d_K = 5\,184\,729 = 2277^2 = 3^4 \cdot 11^2 \cdot 23^2,$$

a monogenic totally real A_5 quintic ramified at three primes rather than one. The Serre obstruction vanishes; the isometry exists with $\det P = 1/2277$, again with every denominator equal to $\sqrt{d_K}$. Over 11 297 primes, Test A is exact ($a_p = 0$ identically on all 2 846 primes of type $(2, 2, 1)$), Test D has worst $|z| = 1.67$ across the nine classes of $\mathrm{SL}_2(\mathbb{F}_5)$, and the Sym^2 and Sym^4 identities hold to 1.1×10^{-16} and 8.9×10^{-16}. Neither the construction nor its validation suite is specific to the 15733 field.

13 Error budget

13.1 Truncation

Since $\det \rho = 1$, each a_{p^k} is a sum of $k + 1$ terms of modulus 1, so $|a_{p^k}| \le d(p^k)$ and $|a_n| \le d(n)$ by multiplicativity. This bound is attained: over 1.2×10^6 coefficients the maximum of $|a_n|/d(n)$ is exactly 1.000000. With $K_0(x) < \sqrt{\pi/2x}\, e^{-x}$ and $d(n) \le 2\sqrt{n}$,

$$|G(v) - G_M(v)| \;\le\; 2\sqrt{\frac{\pi}{2c}} \cdot \frac{e^{-c(M+1)}}{1 - e^{-c}}, \qquad c = \frac{2\pi v}{\sqrt{N}},$$

every step an inequality. At $M = 1.2 \times 10^6$ this is 10^{-203} at $v = 1$ and 10^{-133} at the smallest v used anywhere.

13.2 Quadrature

Evenness of $\tilde{F}$ means the folded half-line sum *with half weight at* $u = 0$ is exactly the full-line trapezoid, and Poisson summation makes the error pure aliasing, $\sum_{m \neq 0} \hat{\tilde{F}}(m/du)$. Since $\hat{\tilde{F}}$ is Λ itself, the error is $|\Lambda|$ at height $2\pi/du$, bounded by the archimedean factor times a convexity bound: 10^{-1053} at $du = 0.004$. Without the half weight the sum is not the full-line trapezoid and the error reverts to $O(du)$, which at $du = 0.006$ is about 2 against a signal of 10^{-6}; this mis-weighting initially produced 5 zeros in place of 35.

13.3 What actually limits the range

$\Lambda(1/2 + it)$ decays like $e^{-\pi t/2}$ but is computed as a sum of $O(1)$ terms, so the absolute rounding floor is fixed while the signal shrinks. In double precision the signal-to-noise ratio is about 10^7 at $t = 10$, 10^4 at $t = 15$, and 4 at $t = 20$. The count at $T = 20$ is stable under refinement, but $T \approx 20$ is the ceiling; extending to $T = 50$ requires shifting the contour to $\operatorname{Im} u$ near $-\pi/2$, where $\tilde{F}$ remains analytic and the cancellation becomes an explicit prefactor, not more coefficients or a finer grid.

13.4 The Turing bound

A zero missed below T_1 lowers the counting function by 1 throughout a window of length H, shifting $\int_{T_1}^{T_1+H} S$ by $-H$. Measured values of $|\int S|/H$ are $0.193, 0.159, 0.188, 0.028$ on windows of length 4 and 0.051 on a window of length 10—a margin of 5 to 20. Converting this into a proof requires an explicit upper bound for $|\int S|$ for a degree-2 L-function of this conductor and archimedean type, which is a theorem ([2], and Trudgian for explicit constants) and not a computation. We do not supply it, and Theorem 1.4 is stated conditionally on it.

14 Limitations

- The absence of a Turing bound is the one gap in the zero count; without it, a zero pair off the critical line or a pole of small residue is constrained but not excluded.
- The conductor determination is numerical. The norm computation proves only that the support is contained in $\{2, 5, 15733\}$; the exponents come from the functional-equation scan.
- Nothing here constructs a Maass form, and nothing here bears on Artin's conjecture in general. The result is of the same kind as [2]: one representation, verified to finite height.
- The completeness of published enumerations of totally real quintic fields below 2×10^7 is used only for context and in no proof. We note that $15733^2 = 2.48 \times 10^8$ exceeds the 10^8 completeness bound advertised for signature $[5, 0]$ [8], so absence of this field from a table would carry no information, and we claim no novelty for the field itself.
- In an auxiliary computation of the degree-3 character χ_3, the prime 2 was mislabelled: the test distinguishing $5A$ from $5B$ compares V with $\pm\sqrt{\operatorname{disc} f}$, and these coincide modulo 2. The correct class is $5B$ by Remark 3.6. This affects one prime in 5.76×10^6 and no conclusion of this paper.

Author contributions

Dula posed the problem, supplied the icosahedral context and the target primes, and executed the large-scale computations. Claude Opus 5 derived the inertia and conjugacy-class results of §3, identified and proved the orientation lemma and the residue-field shortcut, designed the validation suite and the error budget, implemented the search, construction and L-function code, and drafted the manuscript. Grok 4.6 and Gemini 3.8 contributed the synthematic formulation used to pass between resolvents, the validation protocol adopted in §6, the correction that the lift order does not determine the trace sign uniformly, and independent review of intermediate claims. All computational claims were reproduced independently by at least two of the authors.

Data and code availability

The defining polynomial, integral basis, orthonormaliser P, degree-60 model, 60 automorphisms, coefficient files and all scripts are available from the authors. Verification uses PARI/GP 2.15.4 [9]; searches and L-function computations use Python with NumPy, SciPy and SymPy.

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